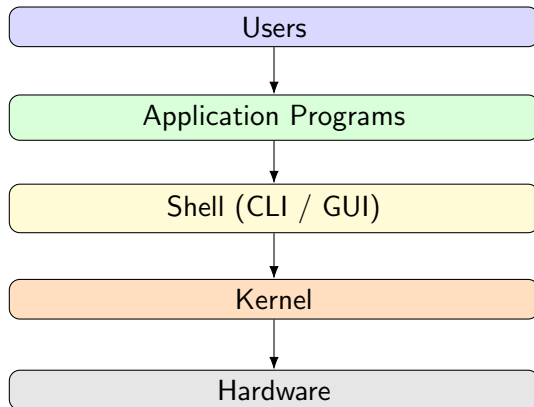


Fundamentals of Operating System Design

Module 2: Shell and Kernel

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DEPSTAR-CHARUSAT

Major Components of an Operating System

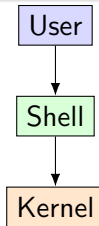


What is a Shell?

Definition

A shell is a command interpreter that acts as an interface between the user and the operating system.

- Accepts user commands
- Interprets commands
- Executes programs
- Displays output



Common Shells

Shell	Command	Platform
Bourne	sh	UNIX
Bash	bash	Linux
C Shell	csh	UNIX
Korn	ksh	UNIX
Z Shell	zsh	Linux/macOS
PowerShell	pwsh	Windows

Linux Demo: `echo $SHELL`
`cat /etc/shells`

CLI

- Keyboard driven
- Fast
- Low memory usage
- Command based

GUI

- Mouse driven
- Easy to learn
- Icons and windows
- User friendly

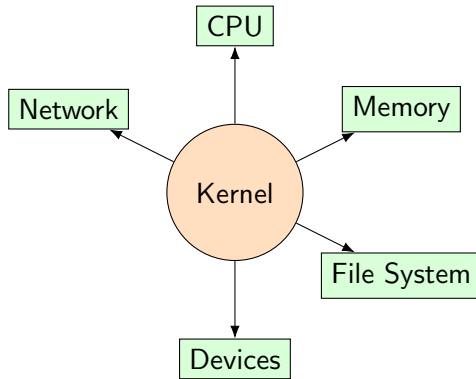
What is a Kernel?

Definition

The kernel is the core of the operating system. It manages hardware resources and provides services to applications.

Kernel = Heart of the Operating System

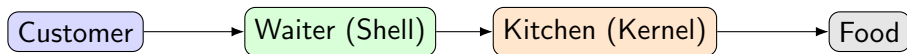
Responsibilities of the Kernel



Shell vs Kernel

Shell	Kernel
User Interface	Core of OS
User Space	Kernel Space
Accepts commands	Controls hardware
Replaceable	Not easily replaceable
Examples: Bash, Zsh	Linux, Windows NT, XNU

Restaurant Analogy



The customer does not enter the kitchen directly. Similarly, the user communicates with hardware through the shell and the kernel.

Summary

- Shell is the user interface.
- Kernel is the core of the OS.
- Kernel manages CPU, memory, files and devices.
- Shell accepts commands and forwards requests.